# FALCUTÉ DES SCIENCES ET TECHNOLOGIES (FST)

## TROISIÈME ANNÉE

Rapport du travail de Laboratoire N$^{0}$ 3

Cours : Réseaux I

Étudiant : NELSON Edwine Ghodson Wilson

Professeur : Ismaël SAINT AMOUR

**LE 8 Novembre 2026**

# Réseaux I

## Adressage IPv4 et IPv6 avec Cisco Packet Tracer

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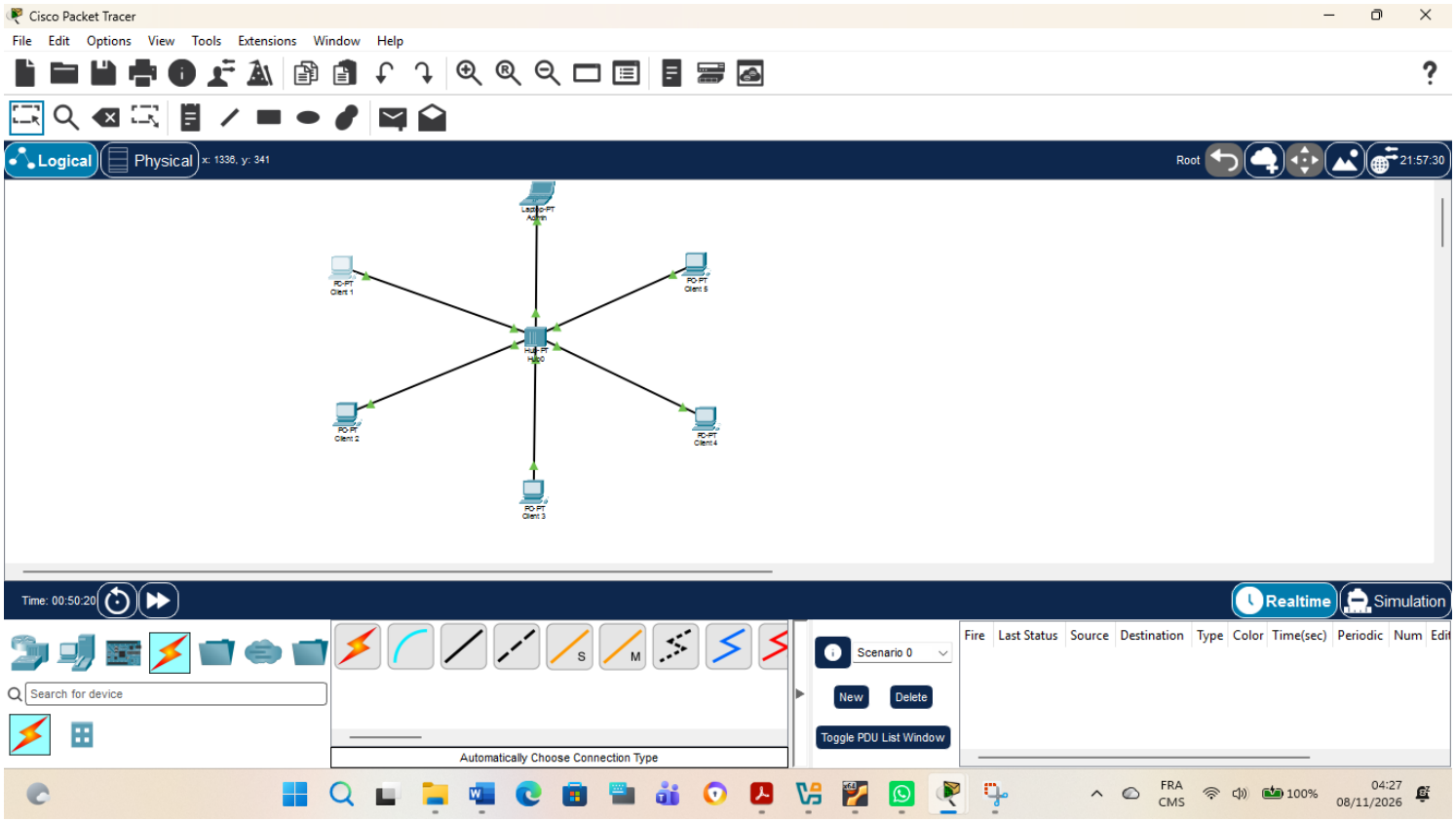
### *TD 3*

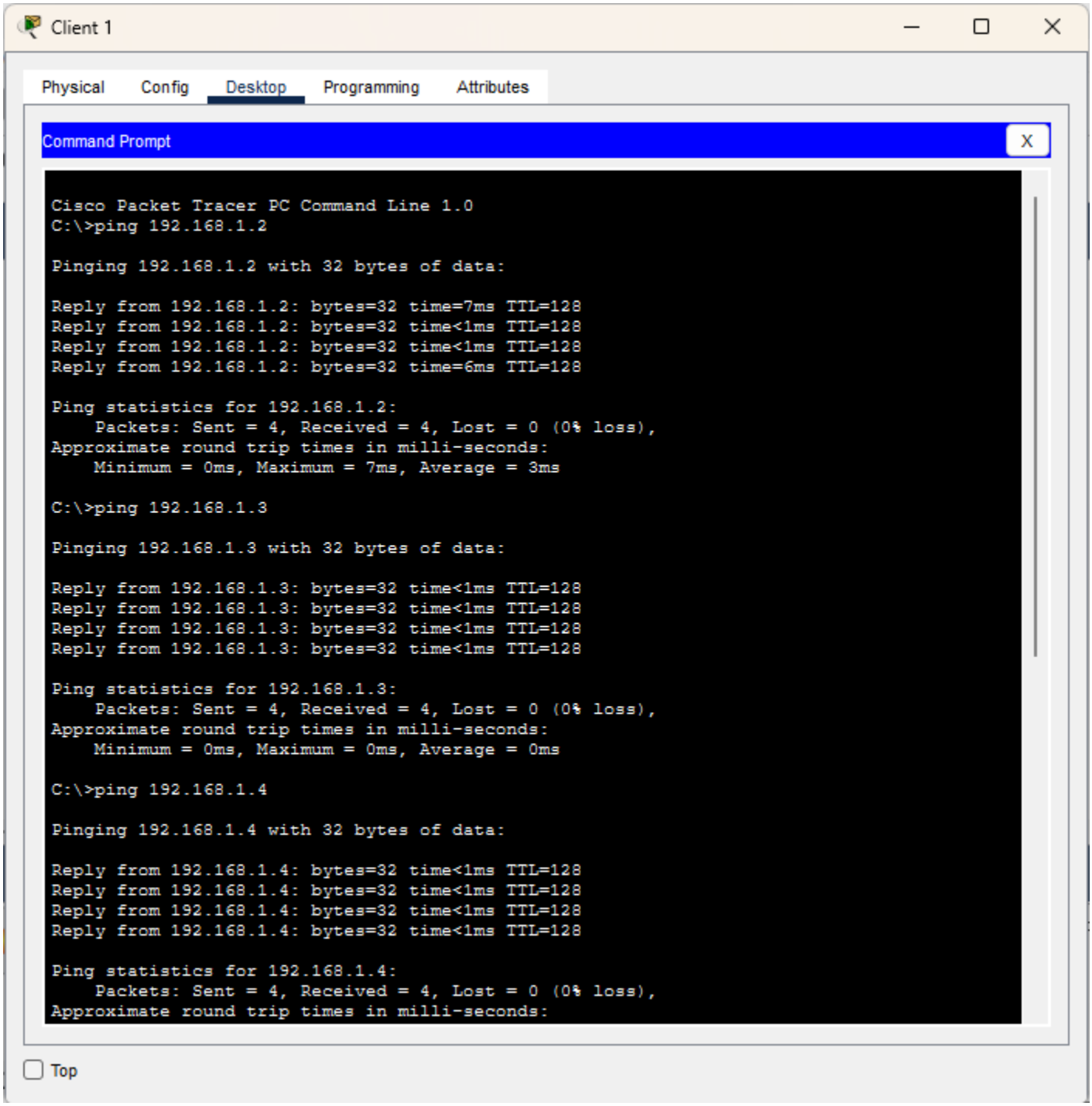
### *L'objectif de ce TD est de :*

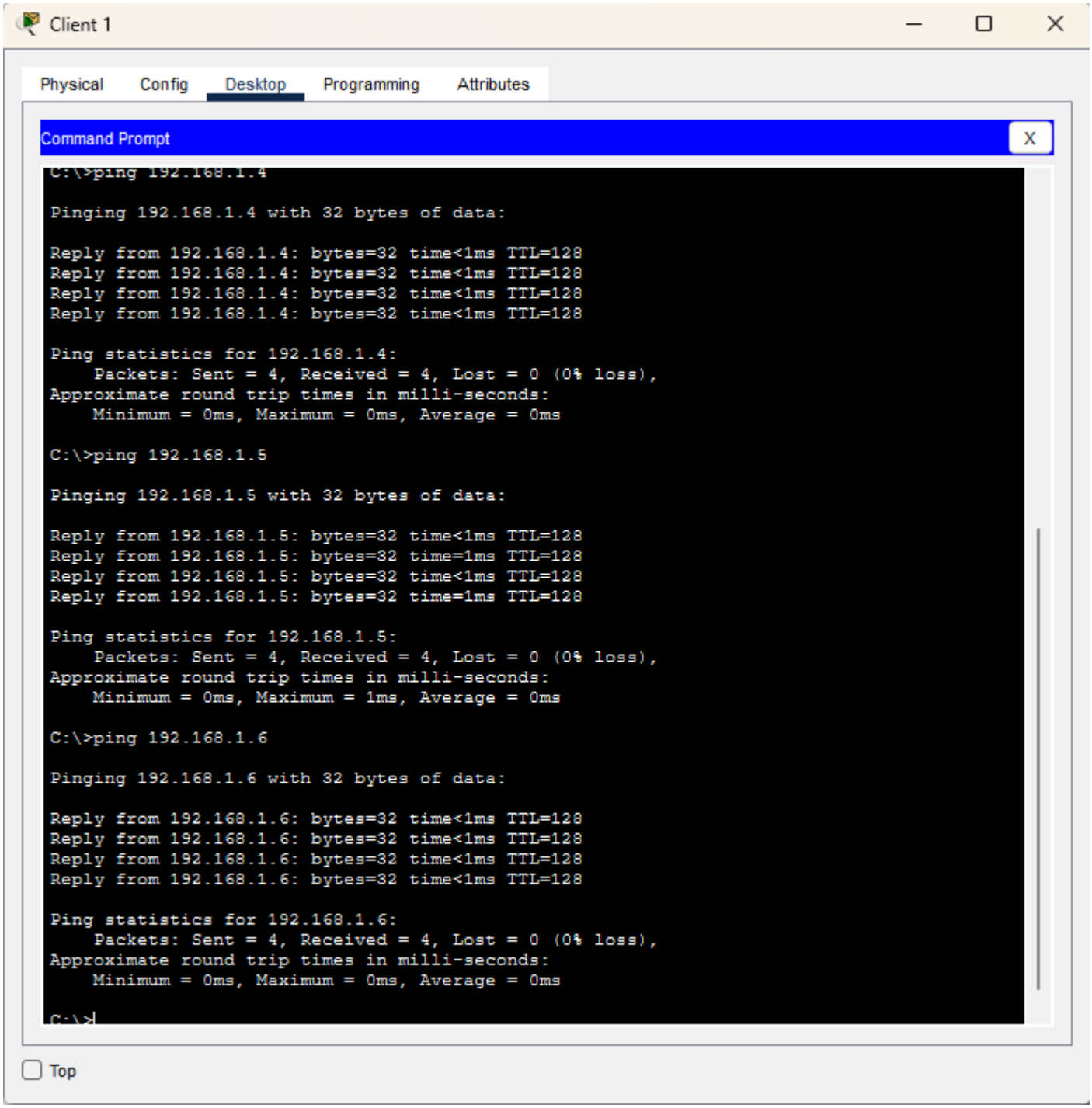
1. Savoir attribuer des adresses IP valides aux machines.
2. Comprendre l'adressage IPv4 et IPv6.
3. Configurer des adresses IP sur des hôtes et routeurs dans Cisco Packet Tracer.
4. Vérifier la connectivité avec les commandes `ping` et `ping ipv6`.

### *Travaux Dirigés*

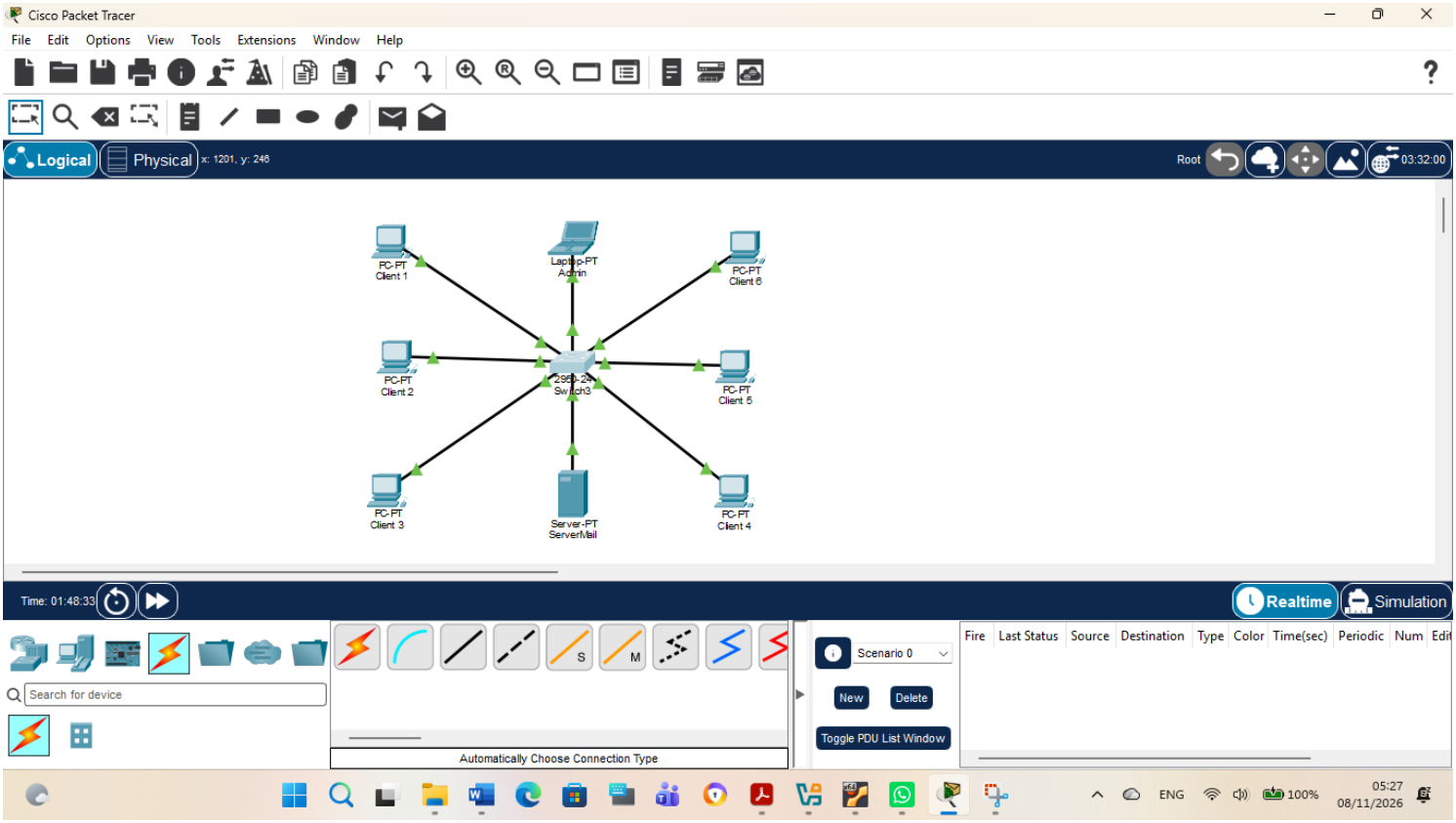
1. Reproduisez cette topologie en configurant les adresses IPv4 indiquées dans le tableau ci-dessous, puis vérifiez la connectivité.

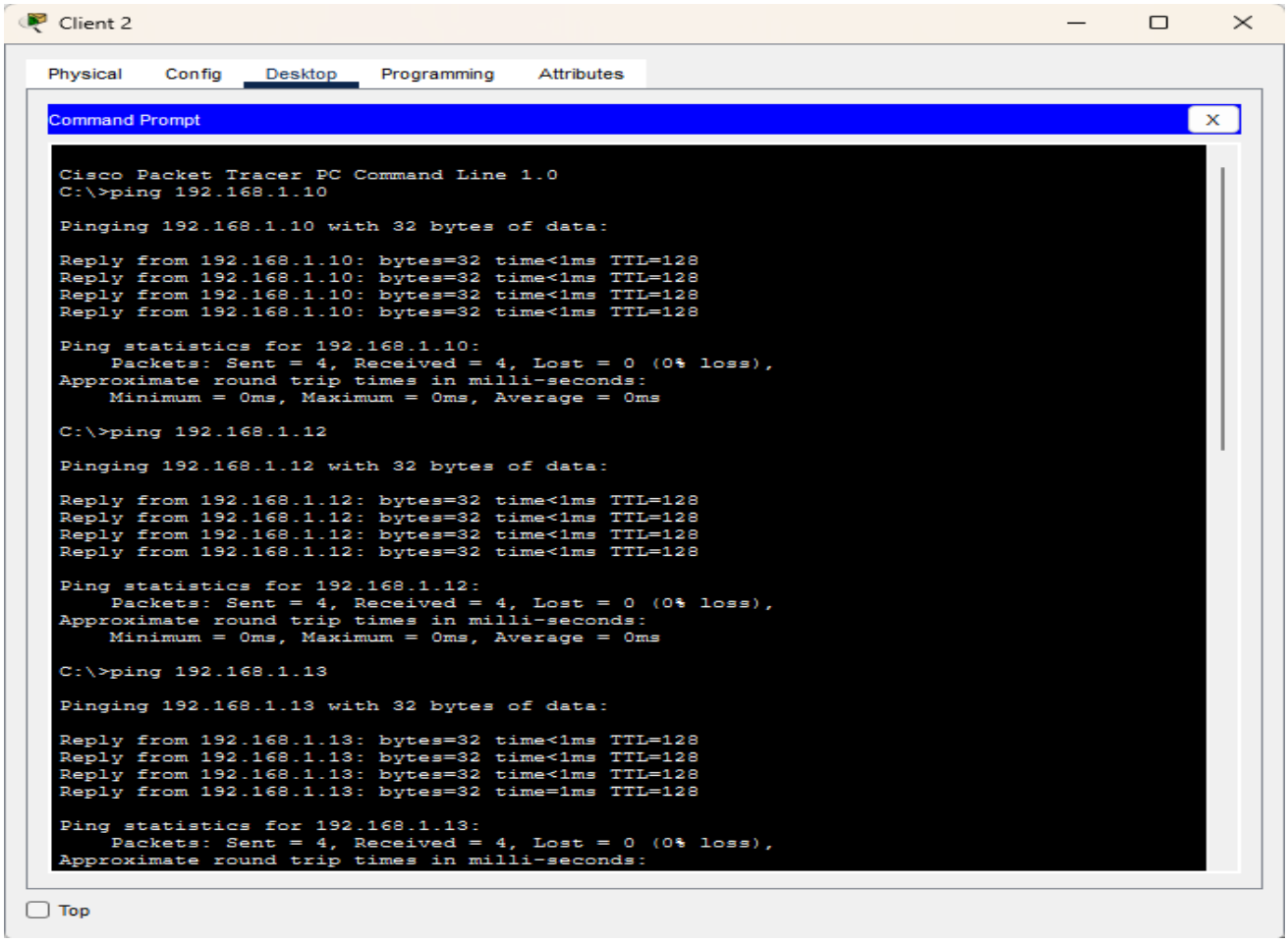


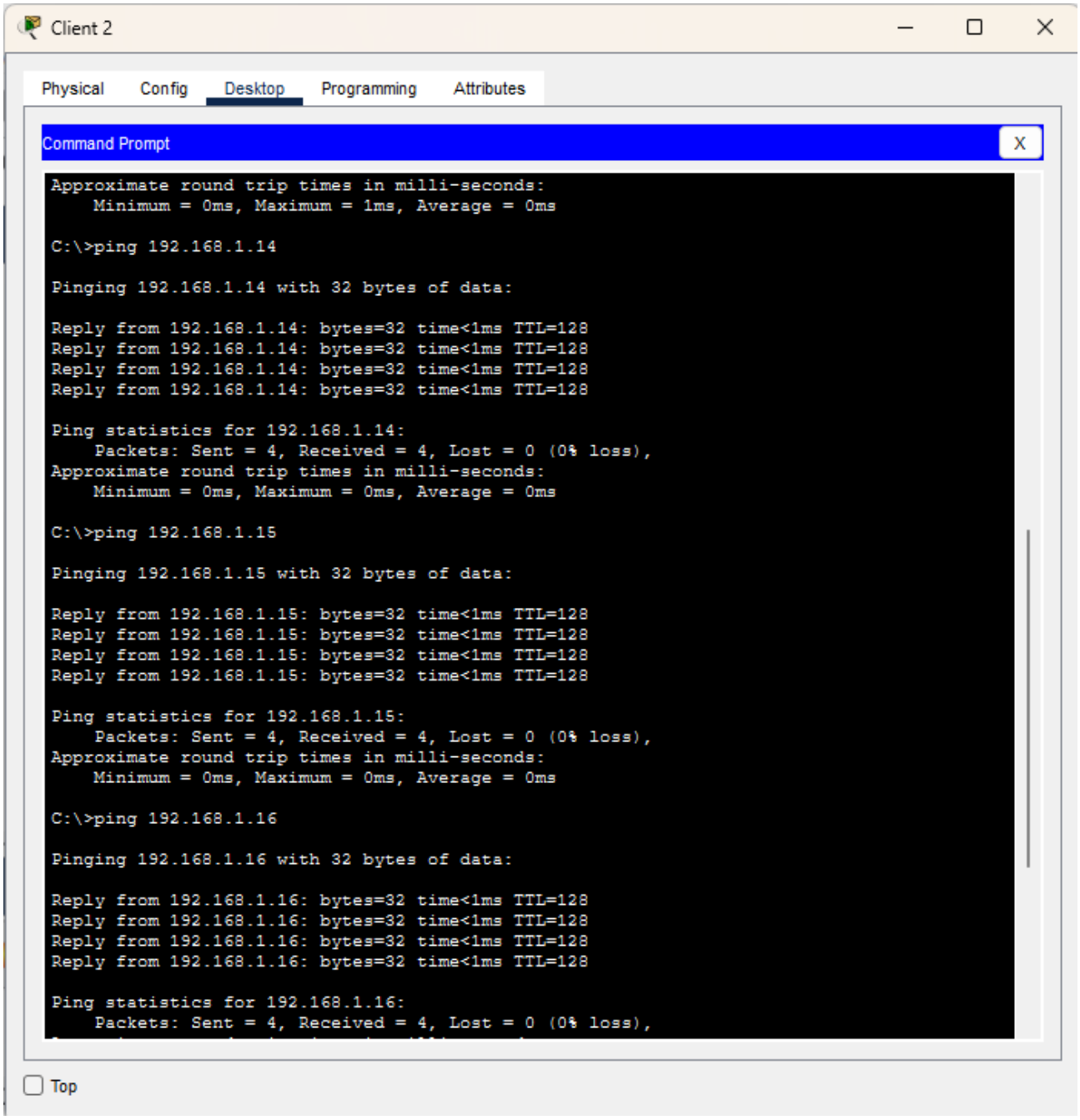

Client 1
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2
Pinging 192.168.1.2 with 32 bytes of data:
Reply from 192.168.1.2: bytes=32 time=7ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=6ms TTL=128
Ping statistics for 192.168.1.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 7ms, Average = 3ms
C:\>ping 192.168.1.3
Pinging 192.168.1.3 with 32 bytes of data:
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 192.168.1.4
Pinging 192.168.1.4 with 32 bytes of data:
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.4:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Top

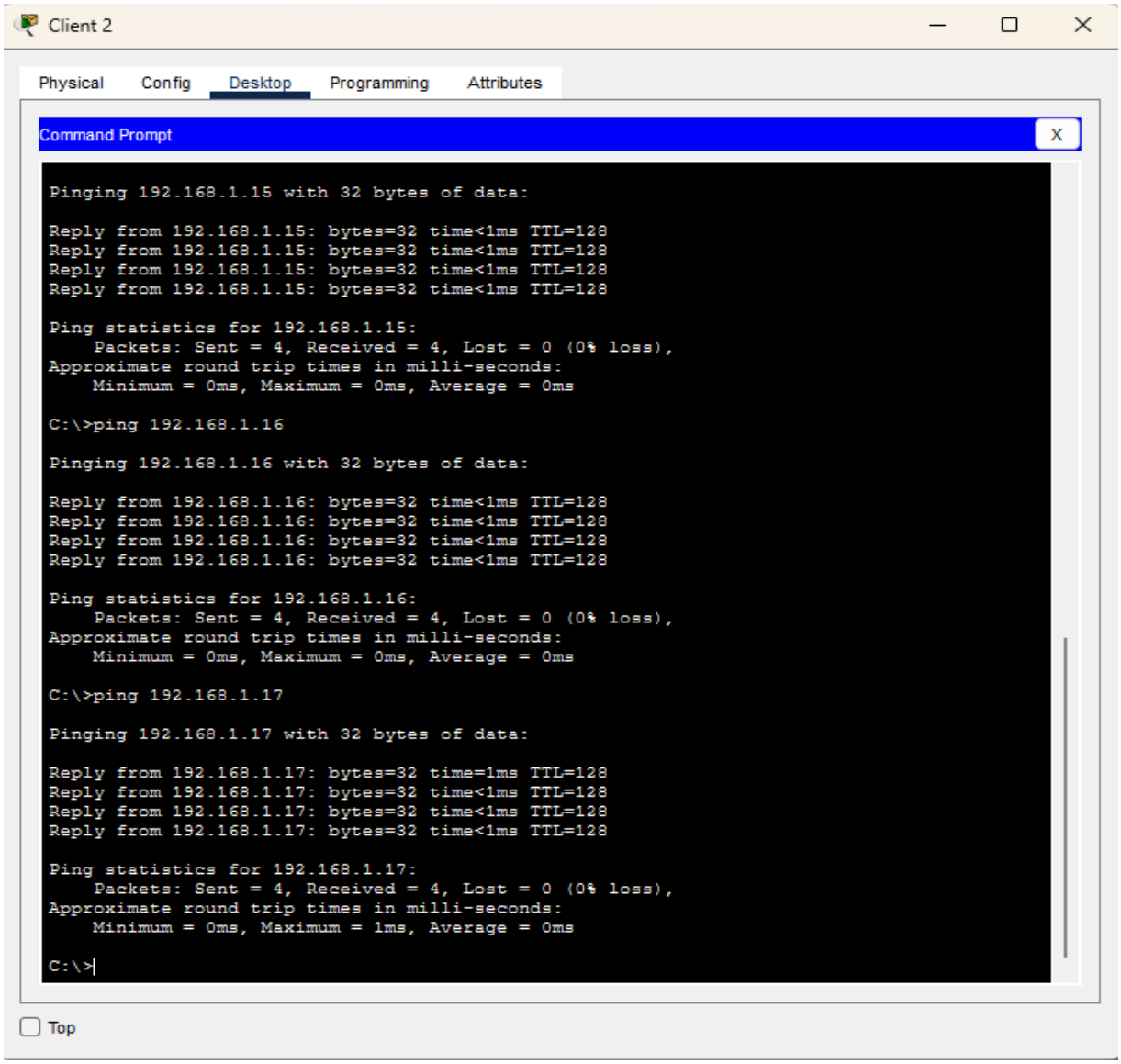



2. Reproduisez cette topologie en configurant les adresses IPv4 indiquées dans le tableau ci-dessous, puis vérifiez la connectivité.

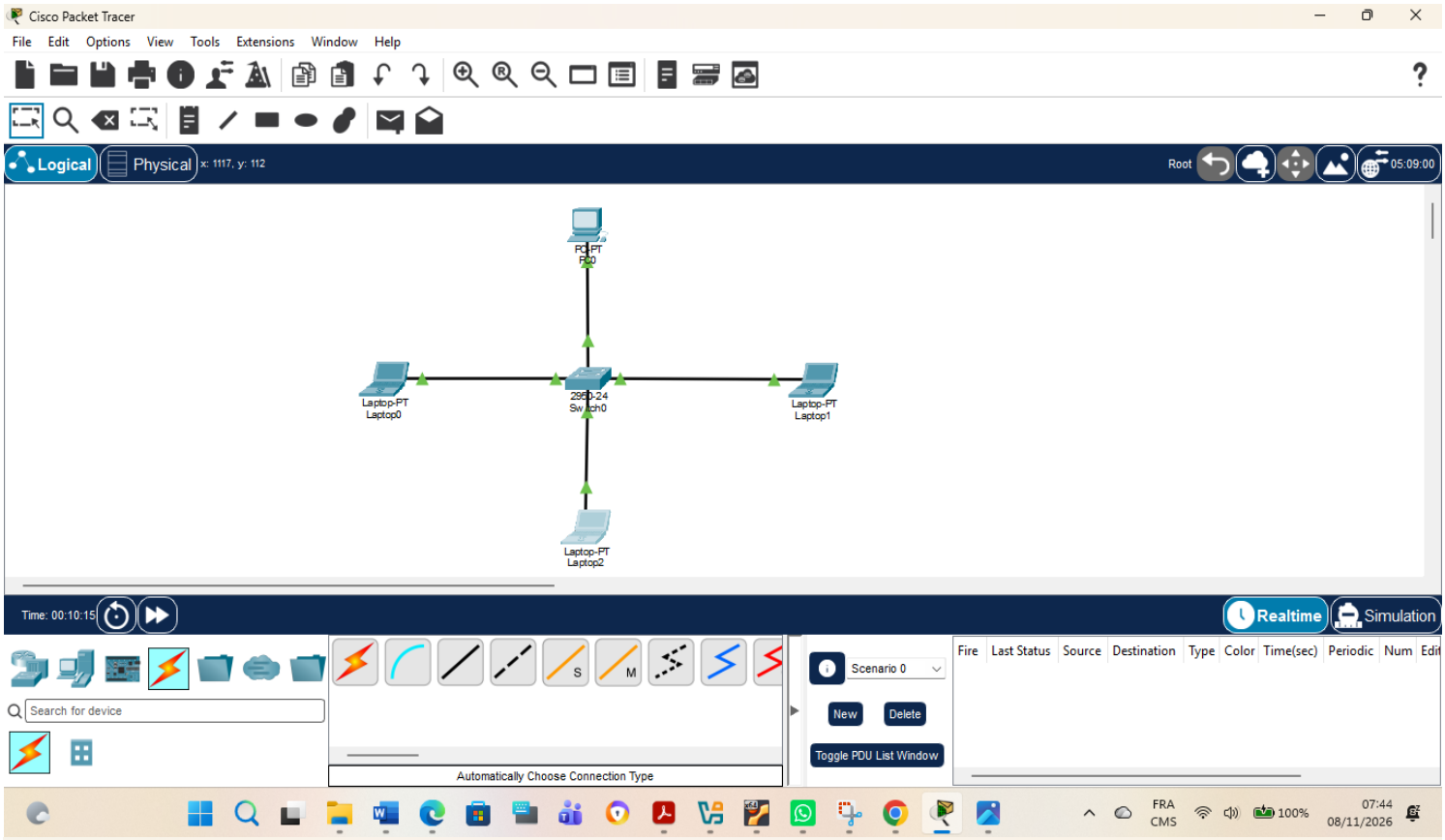

Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 1201, y: 246
Root
03:32:00
PC-PT Client 1
Laptop-PT Admin
PC-PT Client 6
PC-PT Client 2
2960-24 Switch3
PC-PT Client 5
PC-PT Client 3
Server-PT Server/Mail
PC-PT Client 4
Time: 01:48:33
Realtime
Simulation
Search for device
Automatically Choose Connection Type
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
ENG
100%
05:27
08/11/2026

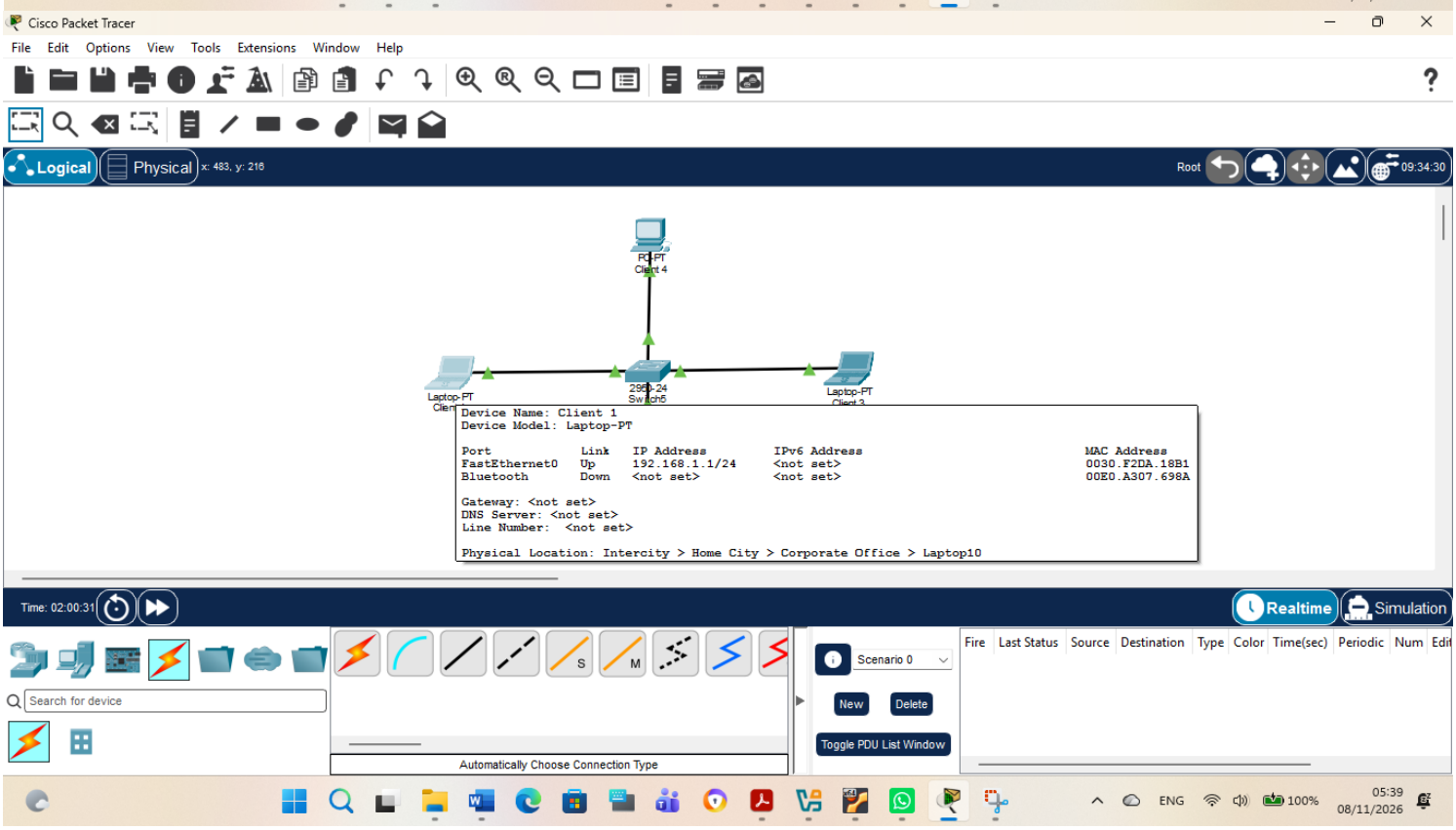

Client 2
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.10
Pinging 192.168.1.10 with 32 bytes of data:
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 192.168.1.12
Pinging 192.168.1.12 with 32 bytes of data:
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.12:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 192.168.1.13
Pinging 192.168.1.13 with 32 bytes of data:
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128
Reply from 192.168.1.13: bytes=32 time=1ms TTL=128
Ping statistics for 192.168.1.13:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Top

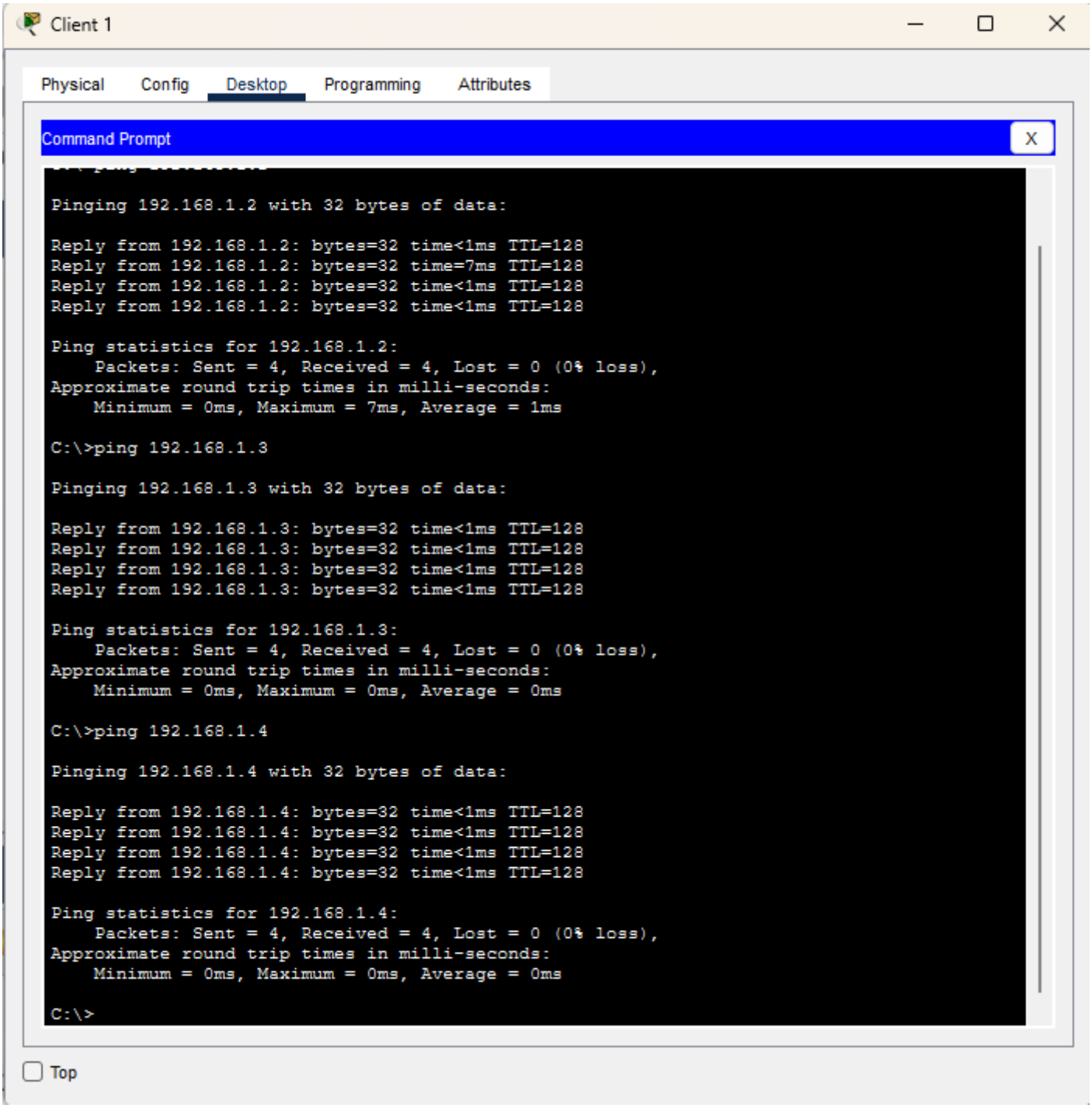

Client 2
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 192.168.1.14
Pinging 192.168.1.14 with 32 bytes of data:
Reply from 192.168.1.14: bytes=32 time<1ms TTL=128
Reply from 192.168.1.14: bytes=32 time<1ms TTL=128
Reply from 192.168.1.14: bytes=32 time<1ms TTL=128
Reply from 192.168.1.14: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.14:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 192.168.1.15
Pinging 192.168.1.15 with 32 bytes of data:
Reply from 192.168.1.15: bytes=32 time<1ms TTL=128
Reply from 192.168.1.15: bytes=32 time<1ms TTL=128
Reply from 192.168.1.15: bytes=32 time<1ms TTL=128
Reply from 192.168.1.15: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.15:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 192.168.1.16
Pinging 192.168.1.16 with 32 bytes of data:
Reply from 192.168.1.16: bytes=32 time<1ms TTL=128
Reply from 192.168.1.16: bytes=32 time<1ms TTL=128
Reply from 192.168.1.16: bytes=32 time<1ms TTL=128
Reply from 192.168.1.16: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.16:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Top



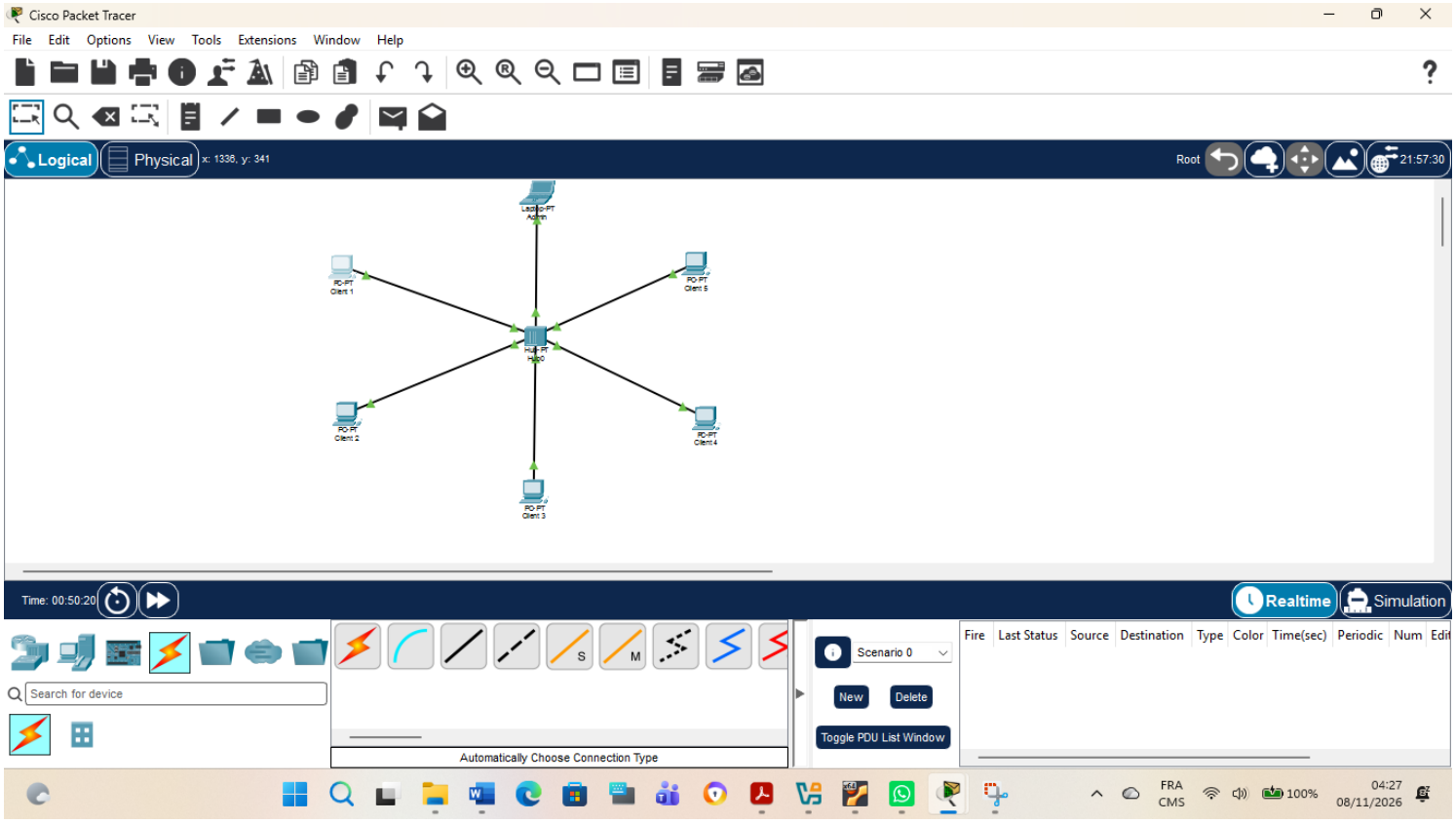

3. Parmi les topologies, choisissez-en une, configurez les adresses IPv4 et testez la connectivité.

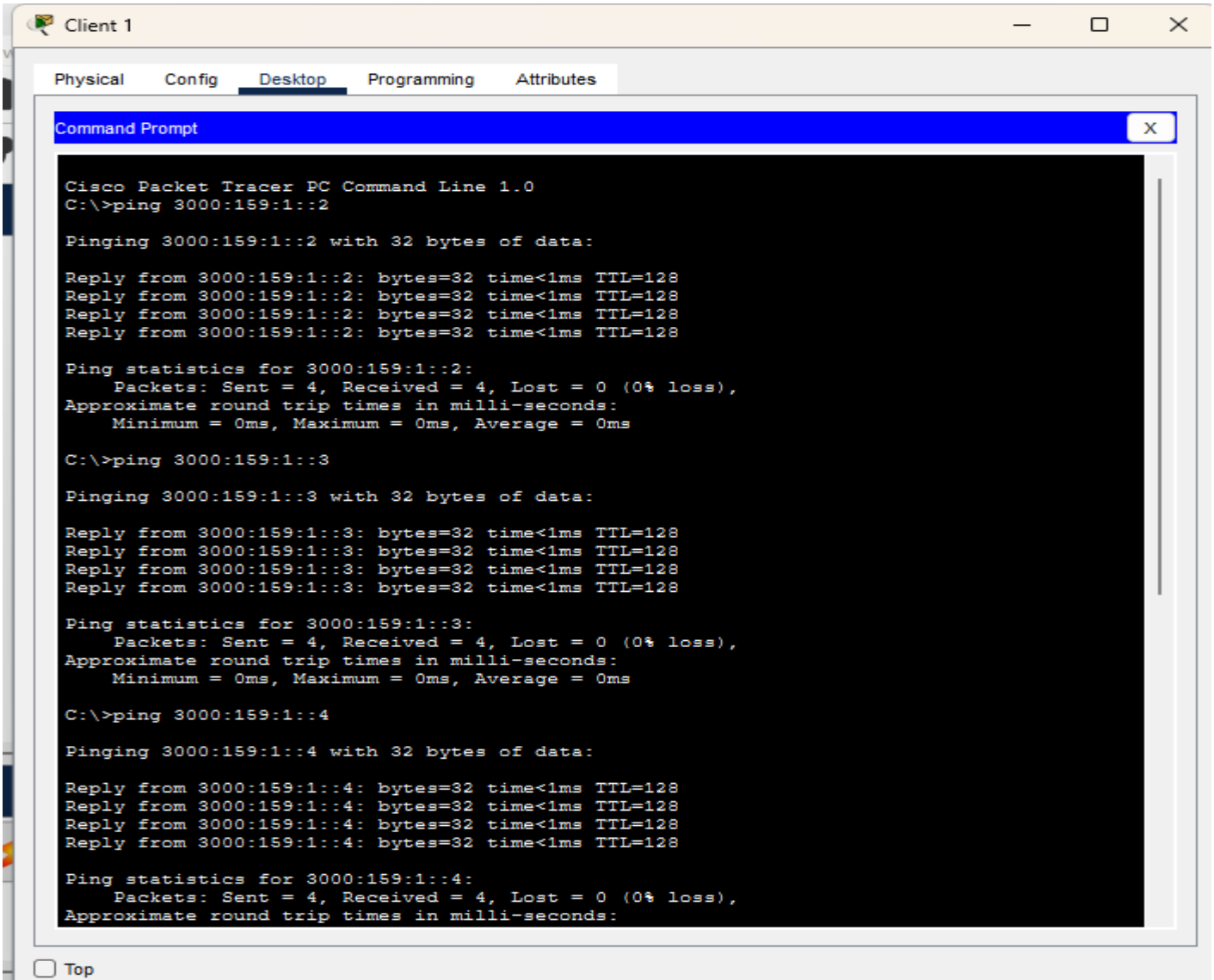

Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical Physical x: 1117, y: 112
Root 05:09:00
PC-PT PC0
Laptop-PT Laptop0
2960-24 Switch0
Laptop-PT Laptop1
Laptop-PT Laptop2
Time: 00:10:15
Realtime Simulation
Search for device
Automatically Choose Connection Type
Scenario 0
New Delete
Toggle PDU List Window
Fire Last Status Source Destination Type Color Time(sec) Periodic Num Edit
FRA CMS 100% 07:44 08/11/2026

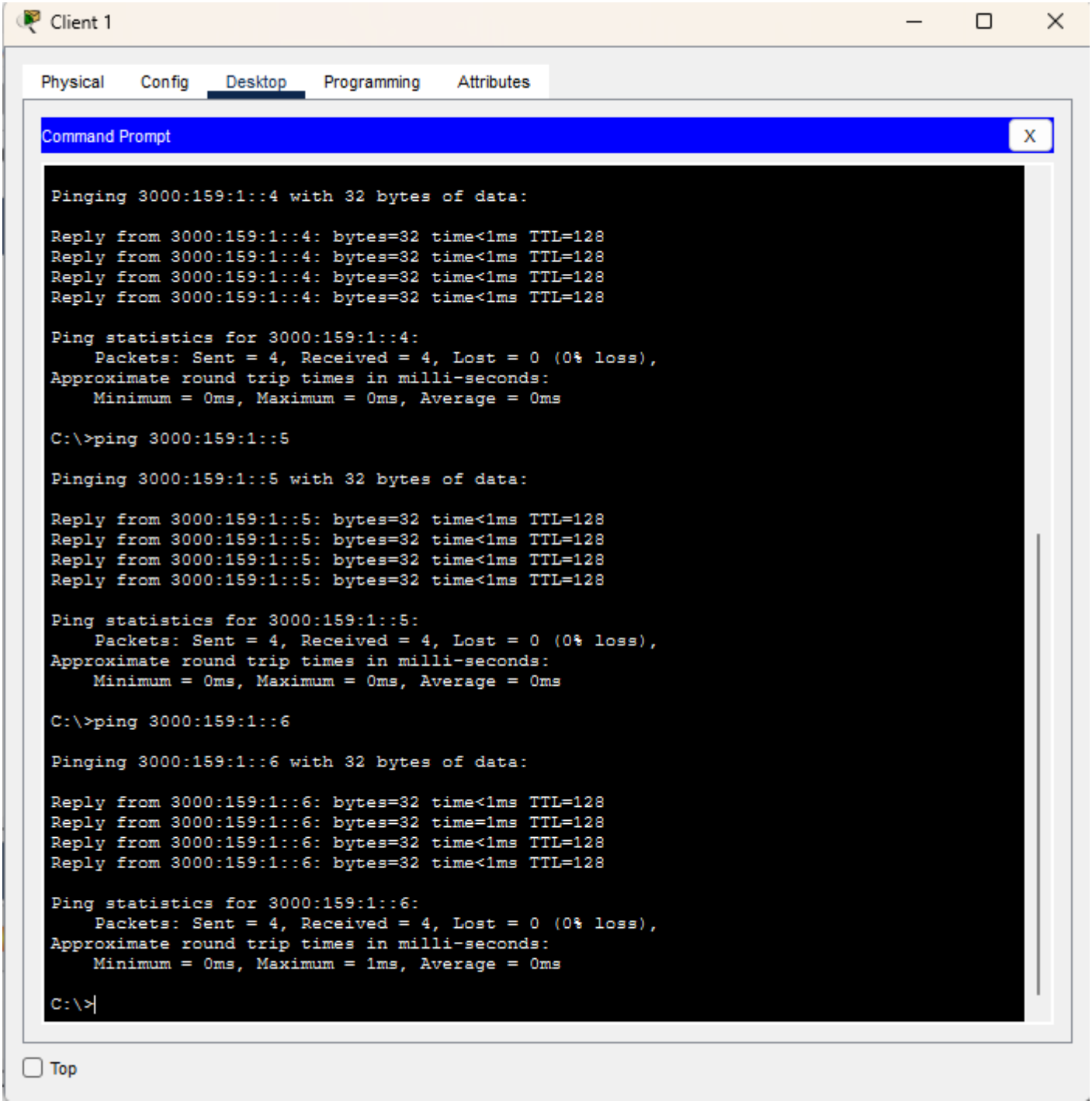

Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical Physical x: 483, y: 216
Root 09:34:30
PC-PT Client 4
Laptop-PT Client 1
2960-24 Switch5
Laptop-PT Client 3
Device Name: Client 1
Device Model: Laptop-PT
Port Link IP Address IPv6 Address MAC Address
FastEthernet0 Up 192.168.1.1/24 <not set> 0030.F2DA.18B1
Bluetooth Down <not set> <not set> 00E0.A307.698A
Gateway: <not set>
DNS Server: <not set>
Line Number: <not set>
Physical Location: Intercity > Home City > Corporate Office > Laptop10
Time: 02:00:31
Realtime Simulation
Search for device
Automatically Choose Connection Type
Scenario 0
New Delete
Toggle PDU List Window
Fire Last Status Source Destination Type Color Time(sec) Periodic Num Edit
ENG 100% 05:39 08/11/2026



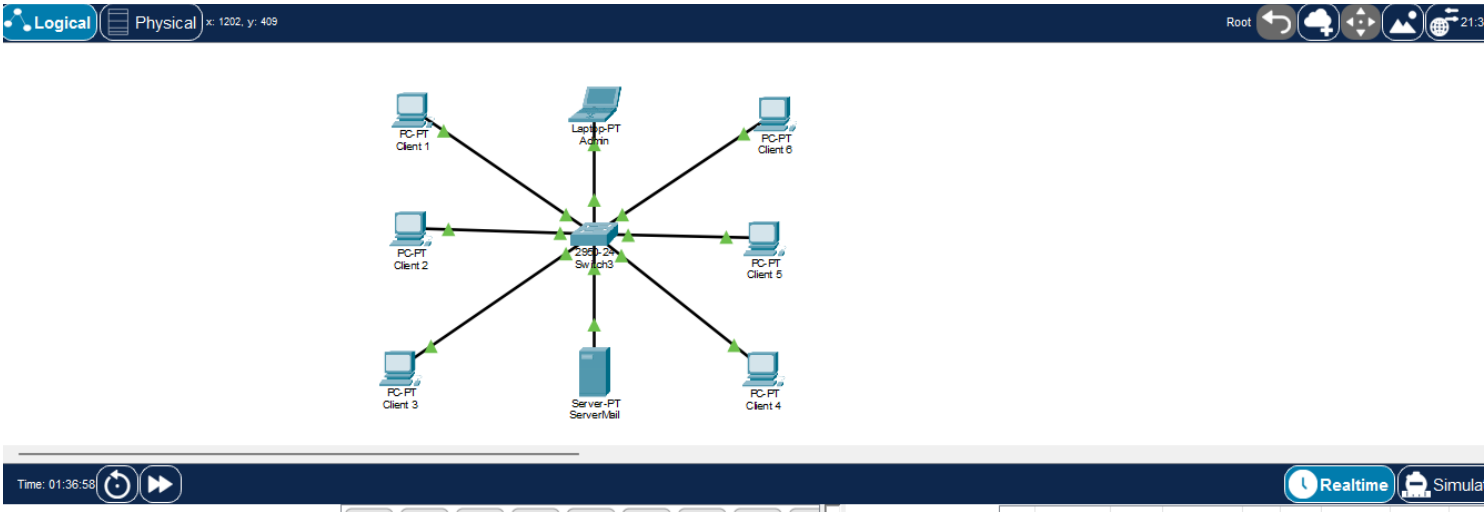

4. Reproduisez cette topologie en configurant les adresses IPv6 indiquées dans le tableau ci-dessous, puis vérifiez la connectivité.


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
Root
Time: 00:50:20
Realtime
Simulation
Search for device
Automatically Choose Connection Type
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
FRA
CMS
100%
04:27
08/11/2026


Client 1
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 3000:159:1::2
Pinging 3000:159:1::2 with 32 bytes of data:
Reply from 3000:159:1::2: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::2: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::2: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::2: bytes=32 time<1ms TTL=128
Ping statistics for 3000:159:1::2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 3000:159:1::3
Pinging 3000:159:1::3 with 32 bytes of data:
Reply from 3000:159:1::3: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::3: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::3: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::3: bytes=32 time<1ms TTL=128
Ping statistics for 3000:159:1::3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 3000:159:1::4
Pinging 3000:159:1::4 with 32 bytes of data:
Reply from 3000:159:1::4: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::4: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::4: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::4: bytes=32 time<1ms TTL=128
Ping statistics for 3000:159:1::4:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Top




5. Reproduisez cette topologie en configurant les adresses IPv6 indiquées dans le tableau ci-dessous, puis vérifiez la connectivité.




Client 1

Physical | Config | Desktop | Programming | Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 3000:159:1::2

Pinging 3000:159:1::2 with 32 bytes of data:

Reply from 3000:159:1::2: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::2: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::2: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::2: bytes=32 time<1ms TTL=128

Ping statistics for 3000:159:1::2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 3000:159:1::3

Pinging 3000:159:1::3 with 32 bytes of data:

Reply from 3000:159:1::3: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::3: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::3: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::3: bytes=32 time<1ms TTL=128

Ping statistics for 3000:159:1::3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

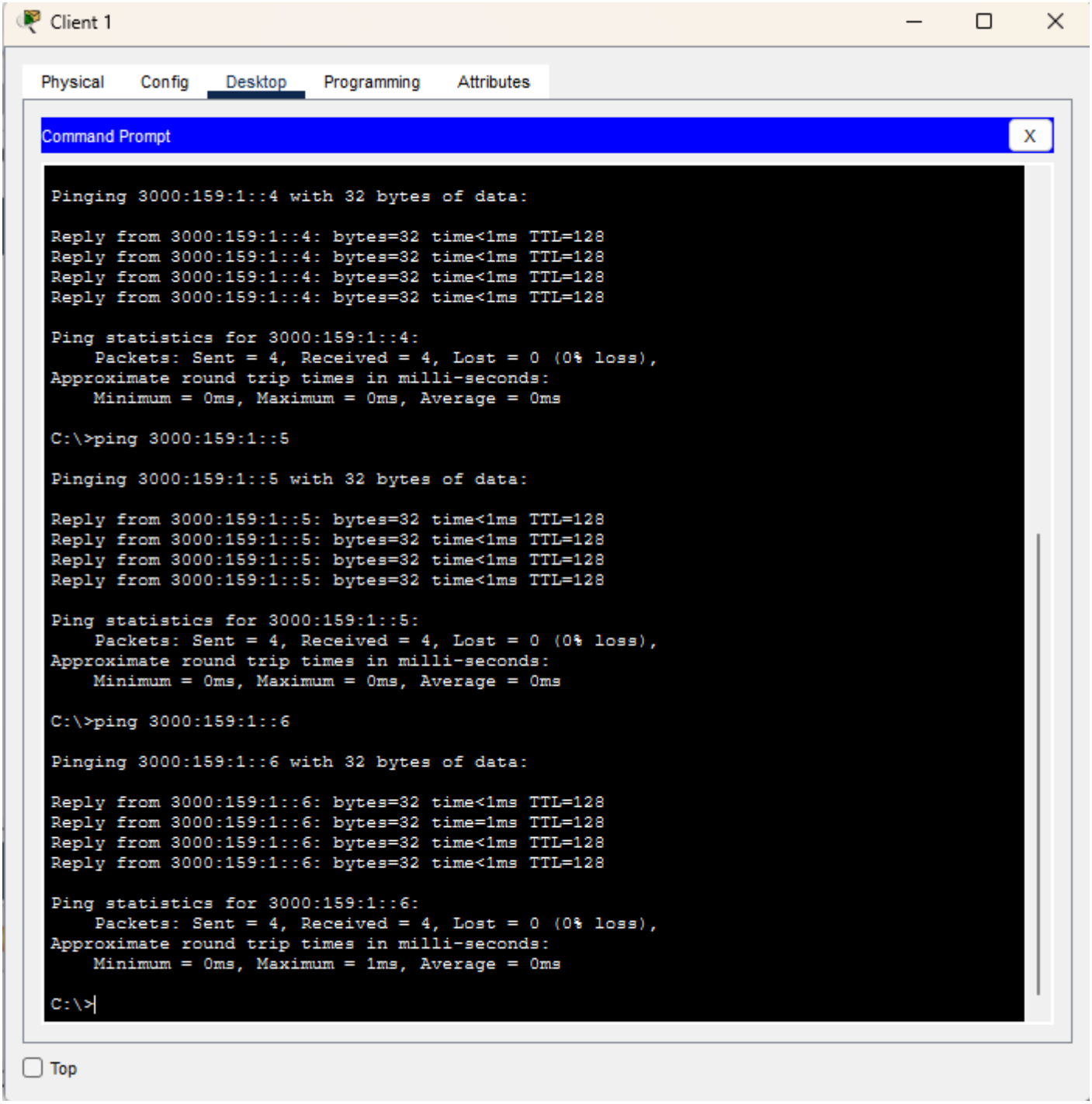
C:\>ping 3000:159:1::4

Pinging 3000:159:1::4 with 32 bytes of data:

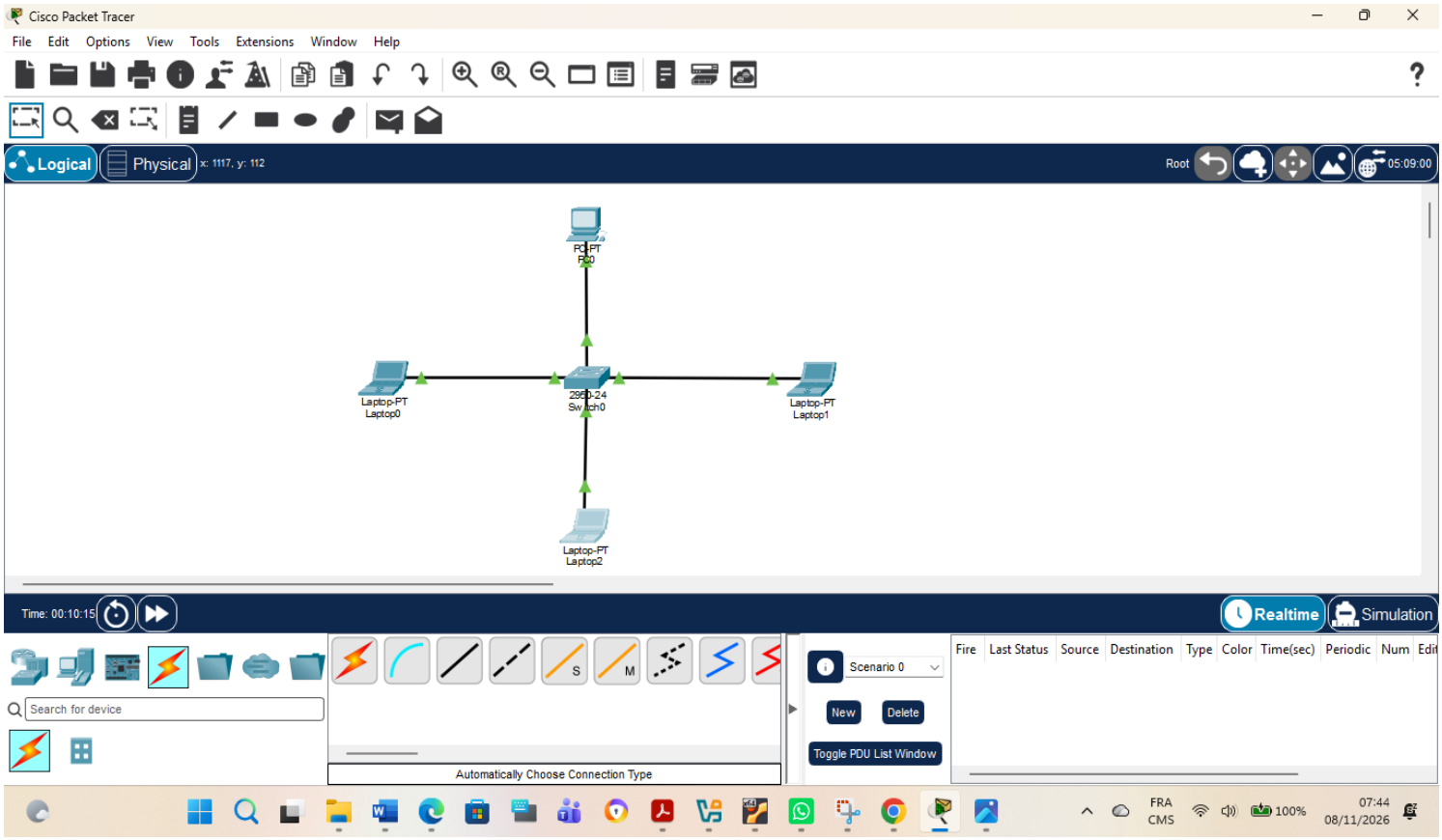
Reply from 3000:159:1::4: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::4: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::4: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::4: bytes=32 time<1ms TTL=128

Ping statistics for 3000:159:1::4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
```

Top




6. Parmi les topologies, choisissez-en une, configurez les adresses IPv6 et testez la connectivité.


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 1117, y: 112
Root
05:09:00
PC-PT PC0
Laptop-PT Laptop0
2960-24 Switch0
Laptop-PT Laptop1
Laptop-PT Laptop2
Time: 00:10:15
Realtime
Simulation
Search for device
Automatically Choose Connection Type
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
FRA CMS
100%
07:44
08/11/2026

| Dispositifs | Ipv4 | Ipv6 |
|---|---|---|
| client 1 | 192.168.1.10 | 3000:is9:1::1 / 64 |
| client 2 | 192.168.1.11 | 3000:is9:1::2 / 64 |
| client 3 | 192.168.1.12 | 3000:is9:1::3 / 64 |
| client 4 | 192.168.1.13 | 3000:is9:1::4 / 64 |
| client 5 | 192.168.1.14 | 3000:is9:1::5 / 64 |
| client 6 | 192.168.1.15 | 3000:is9:1::6 / 64 |
| client 7 | 192.168.1.16 | 3000:is9:1::7 / 64 |
| client 8 | 192.168.1.17 | 3000:is9:1::8 / 64 |

Client 3

Physical Config Desktop Programming Attributes

Command Prompt

```
Pinging 3000:159:1::1 with 32 bytes of data:

Reply from 3000:159:1::1: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::1: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::1: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::1: bytes=32 time<1ms TTL=128

Ping statistics for 3000:159:1::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 3000:159:1::2

Pinging 3000:159:1::2 with 32 bytes of data:

Reply from 3000:159:1::2: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::2: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::2: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::2: bytes=32 time<1ms TTL=128

Ping statistics for 3000:159:1::2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 3000:159:1::4

Pinging 3000:159:1::4 with 32 bytes of data:

Reply from 3000:159:1::4: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::4: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::4: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::4: bytes=32 time<1ms TTL=128

Ping statistics for 3000:159:1::4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Top

## Conclusion

Durant ce TD, j'ai appris à attribuer des adresses IP à des machines, différencier IPv4 et IPv6 et vérifier la connectivité entre les appareils utilisés.